Julia Wang, PhD

jhwang242@gmail.com | (765)-481-9643 | Google Scholar | juliawang22.github.io

PROFESSIONAL PROFILE

Machine learning scientist with degrees in computer science/artificial intelligence (Stanford, BS) and computational neuroscience (Cold Spring Harbor Laboratory, PhD). Passionate about telling stories with data, modeling, and theory to understand how intricate systems work. Over 7 years of machine learning research experience which includes applying deep learning to real-world biological questions, such as drug interaction, sleep signals, multi-omic genomics and imaging data, and more.

TECHNICAL SKILLS

- **AI/ML Tools**: HuggingFace, PyTorch, TensorFlow, Flax/Jax, Weights & Biases, Hyperopt
- **Programming Languages**: Python, C++, C, Java, Matlab, SQL, Arduino, CSS/HTML, git
- **Data science and visualization**: numpy, matplotlib, seaborn, plotly
- **High Performance Computing & Cloud Computing**: Slurm, AWS, Google Cloud
- **Relational Databases**: Snowflake, MySQL, pandas

EDUCATION

PhD	**Cold Spring Harbor Laboratory (CSHL)**, Computational Neuroscience	2023
BS	**Stanford University**, Computer Science (Artificial Intelligence)	2019

EXPERIENCE

NeuroAI Research Scientist| Axoft, Inc. June 2025 -present

- Led research projects on **foundation models** of multimodal neural signals for disorders of **consciousness**.
- Analyzed large scale neural data such as **LFP and spike signals** and developed **reproducible data processing** pipelines.

Neuroscience Research Fellow | Albert Einstein College of Medicine, Bronx, NY 2024-2025

- Training neuroscience inspired AI models of **communication between brain areas**.
- Utilizing mechanistic **interpretation** of **computer vision** models with **low-rank optimization.**

Algorithms Research Intern | Enable Medicine, Menlo Park, CA Summer 2023

- Applied AI methods on **clinical genomics data**, working towards eventual **drug discovery** and patient outcome models at an **early stage startup**
- Developed **statistical and LLM** based algorithms for analyzing single cell and spatial **transcriptomics and proteomics** data.

Computational Neuroscience Researcher (PhD) | Cold Spring Harbor Laboratory 2019-2024

- Invented an **unsupervised AI** method to **neuroscience data** to classify **sleep states**.
- Analyzed **time-series and spatiotemporal data** (EEG, LFP, Calcium Imaging) and theory for **probabilistic deep learning** models
- Managed all SQL-based **databases**, group-wide **codebases**, and GPU **servers**
- **Mentored** summer interns and incoming students and organized coding lectures

Computational Neuroscience Research Intern | Cold Spring Harbor Laboratory Summer 2018

Software Engineering Practicum Intern | Google, Mountain View, CA Summer 2017

- **Systems software engineering** for deep learning **model evaluation** in C++ & Python
- Contributed to a **production-scale end-to-end** machine learning platform in Tensor Flow Extended (TFX).

Computational Neuroscience Research | Stanford University, Stanford, CA. 2017 - 2019

- Trained and evaluated convolutional neural networks (CNNs) of the retina in Tensorflow and PyTorch

PUBLICATIONS AND PRESENTATIONS

Published in peer-reviewed journals and conference proceedings including NeurIPS workshops, *Neuron*, and *Neural Computation*. Presented talks and posters at top neuroscience and machine learning conferences, such as Society for Neuroscience (SFN), Computational and Systems Neuroscience (COSYNE)

Publication Highlights:

- **Wang. JH**, *et al.* A manifold of heterogeneous brain states across cortical areas (*in prep).*
- **Wang, JH**, e*t al.* Variational autoencoder for learning robust representations of time-series data (*NeurIPS*: *UniReps: the First Workshop on Unifying Representations in Neural Models).*
- Others:
- Shen, Y., **Wang, J**. & Navlakha, S. A correspondence between normalization strategies in artificial and biological neural networks. *Neural Computation (*2021) doi:10.1101/2020.07.17.197640
- Maheswaranathan, N. *et al.* Interpreting the retinal neural code for natural scenes: From computations to neurons. *Neuron* (2023)
- **Wang, J.**, Jennings, A. K. & Kowalski, J. R. The Anaphase-Promoting Complex (APC) ubiquitin ligase affects chemosensory behavior in C. elegans. *PeerJ* **4**, e2013 (2016). doi: 10.7717/peerj.2013

Conference presentations:

”A manifold of heterogeneous brain states across cortical areas”

COSYNE ’22(Poster), Society for Neuroscience ’22 (Poster), EAEEG ’23 (talk), COSYNE ’23 (poster)

“Multi-modal, multi-species, and multi-task latent-space model for decoding level of consciousness”

NeurIPS: Foundation Models for the Brain and Body (non-archival), *biorXiv*

LEADERSHIP & TEACHING

Vice President, Diversity Initiative & Advancement for Science | CSHL, *Vice President*

- Organized coffee chats to support underrepresented trainees and create a safe space
- Represented the graduate office and community at HBCU and community college recruitment
- Launched workshops for graduate applications and essays

President, Kids with Dreams | Stanford University

- Led an over 100 person club in planning and managing events for kids with disability, including weekly programs and yearly events such as Winter Formal.
- Liaised with 300+ families in the Bay Area and worked with local nonprofits

Practical Coding Series | CSHL, *Organizer and Lecturer*

Undergraduate Python Course | CSHL, *Organizer and Lecturer*

Introduction to Computing at Stanford | Stanford University, *Lecturer*